
Session 3 (2h) — Orientation & Supports

Core concepts (1h)

- **Why orientation matters** — FDM prints build up layer by layer, so a part's strength is directional (weaker across layer lines than along them), and any face that overhangs empty space needs support underneath it.
- **Overhang angle rule of thumb** — most FDM printers can bridge overhangs up to about **45° from vertical** without support; steeper than that usually needs support material.
- **Support types:**
 - Support everywhere vs “support on build plate only” (avoids supports touching the model on the inside of enclosures, where they're hard to remove)
 - Tree/organic supports — fewer contact points, easier removal, better for organic shapes
 - Grid/normal supports — more predictable for flat overhangs, harder to clean off
- **Orientation tradeoffs to weigh:**
 - Minimize total support material (faster, less waste, less cleanup)
 - Put the strongest layer direction along the axis that will see the most load
 - Keep the largest flat face down for bed adhesion
 - Hide seam lines/support marks on non-visible faces

On-site task link

This directly covers 출력 (printing) decisions the internship expects you to make yourself for each part, not just accept the slicer's default orientation.

Hands-on practice (1h)

1. Take your CAD enclosure's **lid** (or any STL with an overhang) and slice it in its default orientation — check the layer preview for red/orange support material.
2. Rotate the part 90° and re-slice — compare support volume and estimated time.
3. Enable “support on build plate only” and observe how the support structure changes for overhangs on the *inside* of the shape vs the outside.
4. Decide and write down your chosen orientation for the enclosure base and lid, with a one-line reason for each.

Checkpoint

- ☐ Can state the ~45° overhang rule of thumb and why layer adhesion is directional
- ☐ Compared support volume across at least 2 orientations for the same part
- ☐ Chose and justified a final print orientation for your enclosure base and lid